

Budget-aware reliability auditing of DFT force datasets

Physical consistency, error prediction and missed failures

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Abstract

Large density-functional-theory datasets support molecular and materials discovery, but the reliability of individual calculations remains a separate question from surrogate-model accuracy. We formulate reliability auditing as a budgeted search for discrepancies between existing force labels and tightly recomputed references. Physical consistency scores and learned error predictors are compared with explicit accounting for the reference calculations used to train the latter. A retrospective benchmark uses 2,000 public paired configurations from AIMNet2 and Transition1x, with molecular-formula separation between seed references and unseen audit pools. At an average total reference budget of 28.04% of each sample, direct force/torque screening recovers 95.0% of the worst-discrepancy decile on AIMNet2, compared with 77.6% for either ridge or a spectral bulk-pooled ridge predictor. On Transition1x, the physical screen recovers 31.8%, while a tree predictor reaches 39.0% with substantial scenario variation. More seriously, the physical screen misses the single configuration contributing 97.75% of the sample’s squared discrepancy. Elementary bounds explain why small symmetry violations cannot certify accurate forces, and a decision formulation distinguishes error-count from error-impact objectives. The results support budget-aware, multi-indicator auditing while providing no evidence for universal detection or a special random-matrix advantage.

1 Introduction

The usefulness of a computational dataset depends on the reliability of its labels, not only on its size. A model can agree with held-out DFT labels and still inherit their numerical or protocol-dependent errors. For a discovery workflow, the relevant question is twofold: how accurately the surrogate reproduces a reference calculation, and whether that calculation is sufficiently reliable for the intended decision. The present work addresses the second question through force-data auditing.

DFT verification is an established subject. [Carbogno et al. \[2022\]](#) examined numerical errors across electronic-structure codes and proposed a basis-set error model for heterogeneous materials databases. [Bosoni et al. \[2024\]](#) developed reproducible cross-code workflows and a broad reference set for testing computational precision. More directly, [Kuryla et al. \[2025\]](#) compared public force datasets with tightly recomputed values and documented appreciable discrepancies. Targeted data audits are also being studied in materials informatics: [Mahshook and Banerjee \[2026\]](#) combined model explanations with first-principles verification to investigate spin Hall conductivity predictions and training labels. These precedents rule out treating the general need for DFT data-quality assessment as an unexplored idea.

Our question is operational: *given an existing dataset and a limited reference-calculation budget, which records should be recomputed?* A learned error predictor can appear attractive when its training references are treated as free. A fair deployment comparison must include them in the budget and allow elementary screening rules to spend the same total number of references

directly. A rule that detects many moderately inaccurate records also need not identify the few records dominating total squared error.

We provide a mathematical statement of these distinctions and a retrospective paired-data benchmark. The method classes are established tools, including extremely randomized trees [Geurts et al., 2006] and covariance regularization motivated by random matrix theory (RMT) [Marchenko and Pastur, 1967, Donoho et al., 2017]. The contribution examined is the combination of physical identifiability limits, explicit seed-cost accounting and detection-versus-impact comparisons. We do not claim a new universal noise estimator, discovery of the source data’s discrepancies, or priority for the general concept of active data auditing.

2 What does reliability mean?

For a fixed geometry and a specified electronic-structure protocol M , a useful bookkeeping decomposition for a computed property is

$$y_{\text{reported}} = y_{\text{physical}} + b_M + \delta_{\text{numerical}} + \delta_{\text{protocol}}. \quad (1)$$

Here b_M denotes approximation relative to the intended physical quantity, while the remaining terms describe numerical and protocol discrepancies. This is a conceptual decomposition: a single reported value does not identify its terms. Repeated use of the same functional does not average away its systematic bias, and nominally similar calculations may converge to different electronic solutions. Statistical noise need not be independent, Gaussian or zero-mean.

We use a tight recalculation at the same nominal level of theory as an *operational reference*. For configuration i , with n_i atoms, define

$$e_i = F_i^{\text{reported}} - F_i^{\text{ref}}, \quad q_i = \sqrt{\frac{\|e_i\|^2}{3n_i}}, \quad w_i = \|e_i\|^2. \quad (2)$$

The frame RMSE q_i defines high-discrepancy records; w_i measures contribution to pooled squared force discrepancy. Agreement with this reference is not agreement with exact quantum mechanics or experiment. The comparison also does not uniquely identify the numerical cause of every mismatch.

An audit should distinguish identifying records exceeding a tolerance, estimating the dataset’s discrepancy distribution, and checking whether discrepancies change a discovery decision. The experiment evaluates the first task and an aggregate impact measure. It does not establish calibrated per-record confidence intervals or improvements in materials ranking, binding affinity or trained-potential accuracy.

3 Physical checks cannot certify all force components

For an isolated molecule, let A_R map a stacked force vector to net force and torque about the centered geometry:

$$A_R F = \begin{pmatrix} \sum_j F_j \\ \sum_j (r_j - \bar{r}) \times F_j \end{pmatrix}. \quad (3)$$

Let U_R be an orthonormal basis for $\text{range}(A_R^T)$, with actual rank retained for degenerate geometries. Define $Q_R = U_R U_R^T$ and $P_R = I - Q_R$. Our scores include net-force magnitude per atom and rigid-component RMS:

$$s_{\text{net}} = \frac{\|\sum_j F_j\|}{n}, \quad s_{\text{rigid}} = \frac{\|Q_R F\|}{\sqrt{3n}}. \quad (4)$$

Proposition 1 (A one-sided consistency check). *If $A_R F_* = 0$, then*

$$\frac{\|F - F_*\|^2}{3n} = \frac{\|P_R(F - F_*)\|^2}{3n} + s_{\text{rigid}}^2. \quad (5)$$

There is no finite upper bound on total force error determined solely by s_{rigid} when the allowed subspace contains a nonzero vector.

Proof. Orthogonality gives the decomposition and $Q_R F_* = 0$ fixes its normal component. For a unit vector $v \in \ker A_R$, replacing an admissible reference F_* by $F_* + tv$ leaves all constraints satisfied while discrepancy from a fixed reported F becomes arbitrarily large as $|t|$ increases. $\square$

A large rigid component certifies a minimum discrepancy under these assumptions, but a small one cannot certify accuracy. This does not assume every symmetry-valid vector field is conservative. If a numerical reference has residual rigid component $\eta = \|Q_R F^{\text{ref}}\|$, the lower bound becomes $\max(0, \|Q_R F\| - \eta)/\sqrt{3n}$. Such reference residuals are present, though small, in the paired files.

The same geometry limits label cleaning. For an invariant energy model with $F_\theta = -\nabla_R E_\theta$ and $P_R F_\theta = F_\theta$,

$$\|F_\theta - F\|^2 = \|F_\theta - P_R F\|^2 + \|Q_R F\|^2. \quad (6)$$

Orthogonal repair changes a squared force-loss objective only by a parameter-independent constant. Auditing, correcting labels and improving a learner are distinct achievements. Energy-based construction of conservative force models is itself established [Chmiela et al., 2017].

4 Auditing as a budgeted decision

Let $\mathcal{S}$ be the seed set with reference recalculations and $\mathcal{U}$ the remaining records. Let $a_i \in \{0, 1\}$ indicate a new audit, c_i its cost and B the total budget. For tolerance τ , define $b_i = \mathbf{1}\{q_i > \tau\}$ and $p_i = \Pr(b_i = 1 \mid \mathcal{I})$, where $\mathcal{I}$ contains original descriptors and seed reference results. The expected-count acquisition problem is

$$\max_{a_i \in \{0, 1\}} \sum_{i \in \mathcal{U}} a_i p_i \quad \text{subject to} \quad \sum_{i \in \mathcal{S}} c_i + \sum_{i \in \mathcal{U}} a_i c_i \leq B. \quad (7)$$

For an impact objective, replace p_i by $\mathbb{E}[w_i \mid \mathcal{I}]$. Neither objective generally equals ranking by predicted mean log RMSE. Our regressors provide practical risk-ranking scores, not calibrated probabilities or guaranteed solutions to (7).

Proposition 2 (Equal-cost acquisition and score error). *With equal costs and room for k additional audits, selecting the k largest p_i maximizes expected discoveries. If $|\hat{p}_i - p_i| \leq \varepsilon$ for all unaudited records, selecting the k largest $\hat{p}_i$ has expected-count regret at most $2k\varepsilon$.*

Proof. Linearity of expectation makes utility a sum; exchanging a selected smaller p_i with an unselected larger value improves it. For optimal set T_* and estimated-score set $\hat{T}$, insert and subtract estimated scores in $\sum_{T_*} p_i - \sum_{\hat{T}} p_i$. The estimated-score difference is nonpositive by optimality of $\hat{T}$, and each remaining sum is bounded by $k\varepsilon$. Independence between record errors is unnecessary. $\square$

These standard additive-utility arguments specify the objective. The experiment does not establish the uniform probability-error premise. With unequal costs, the binary problem is a knapsack problem; sorting p_i/c_i is not generally an exact solution. Our benchmark counts references equally because individual calculation timing is unavailable. Count savings are not measured CPU-hour savings.

5 Retrospective benchmark

5.1 Paired data and chronology

We use the AIMNet2 and Transition1x paired files released with [Kuryla et al. \[2025\]](#). Each contains 1,000 configurations. Their stored force columns reproduce pooled discrepancies of 4.7979 and 26.9007 meV/Å, respectively. The upstream references use ORCA 6.0.1 with tight grids and RIJCOSX disabled. Files are pinned to the immutable commit

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No new DFT calculations were performed: an audit is simulated by revealing an existing paired reference only when permitted by the split or policy.

The protocol was recorded after a previous force-denoising pilot on these samples, but before fitting the present predictors. These are therefore not fresh independent confirmation data for choices motivated by the earlier analysis. ANI-1x and SPICE files are excluded because attempted unit/dispersion conventions did not reproduce their reported comparisons; Appendix A records the unresolved checks.

5.2 Seed costs and held-out formulas

Complete molecular formulas are assigned to five deterministic hash folds. In each scenario, one fold supplies seed references and the other four form the unseen pool. There are 977 formula groups for AIMNet2 and 122 for Transition1x. Seed counts are 169–224 and 154–276, respectively, averaging 200 in both samples. No formula appears in both seeds and the corresponding audit pool. This excludes fixed-composition conformer leakage but is not a scaffold or reaction-family split. The five scenarios overlap and are not independent datasets.

We acquire the top 1, 2, 5, 10, 20, 30 or 50% of the unseen pool, rounding upward. The primary acquisition fraction is 10%. Seed references count toward both total budget and discovered high-discrepancy records. A direct policy without training references selects from the entire original sample using the same total count. At the primary setting, the average total budget is 28.04%, with ranges 25.3–30.2% for AIMNet2 and 23.9–34.9% for Transition1x. This is not a 10% total-budget experiment.

We also report comparisons conditional on a common already-audited seed set. They answer an operational question when references already exist, but do not establish an advantage for a new campaign that must obtain them first.

5.3 Descriptors and policies

The 38 descriptors use original forces, geometry, element fractions and charge. They comprise eight element fractions; log atom count and charge; transformed net-force, rigid-force, rotational-force and original force RMS; and invariant summaries of force magnitudes, centered radii, pair distances and nearest-neighbor distances. Tight forces, tight energies and derived discrepancies are absent from inputs. SCF histories, grid metadata and electronic-state diagnostics are unavailable.

Direct policies rank by net force, rigid force, original force magnitude or a standardized geometric-novelty score. The last uses unlabeled geometry/composition descriptors from the whole sample, a transductive preprocessing step. Uniform selection has exactly calculable expected recall and impact. A reference oracle ranks by actual frame discrepancy as an unattainable comparator; it is not necessarily optimal for squared-impact utility.

Learned policies include 256-tree ExtraTrees regressors [[Geurts et al., 2006](#)] on raw and log RMSE, a two-physical-score tree ablation, and ridge regression on log RMSE. Tree settings are fixed before evaluation. Ridge penalties are selected by three-fold grouped validation within each seed set and refitted on all seeds. No audit-pool reference enters fitting or penalty selection.

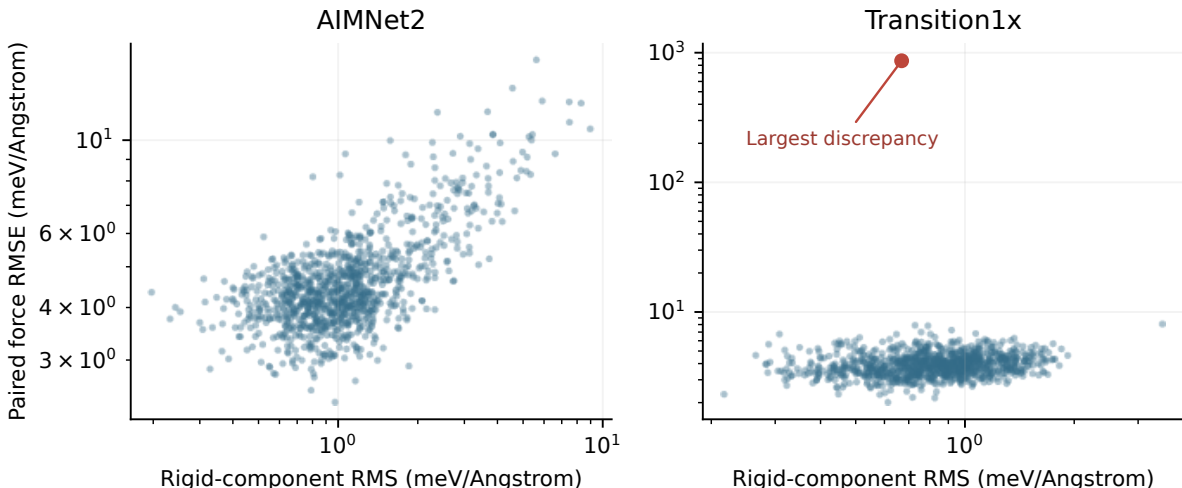


Figure 1: Original rigid-component RMS versus paired discrepancy. The Transition1x record with the largest discrepancy has a modest consistency score. Axes are logarithmic; each point is a configuration.

An RMT-inspired variant pools small eigenvalues of the standardized descriptor Gram matrix before ridge fitting. Its nominal bulk edge is $(1 + \sqrt{p/m})^2$, with p nonconstant descriptors and m seed records. This is a regularization heuristic on correlated descriptors, not an established decomposition of physical force noise. MP assumptions are not verified for these descriptors; covariance estimation theory [Marchenko and Pastur, 1967, Donoho et al., 2017] alone does not imply better auditing.

5.4 Endpoints

The primary high-discrepancy set is the worst 100 records in each sample ranked by reference RMSE. This relative top-decile definition is evaluation truth, not a target used to generate scores or a universal chemical tolerance. Fixed tolerances of 5, 10 and 25 meV/Å are also evaluated. For the set T of seed-plus-acquired records, report

$$\text{recall}(T) = \frac{\sum_{i \in T} b_i}{\sum_i b_i}, \quad \text{impact}(T) = \frac{\sum_{i \in T} w_i}{\sum_i w_i}. \quad (8)$$

All outliers remain in the primary analysis. Scenario means and ranges are reported without treating the overlapping scenarios as independent replications.

6 Results

6.1 Strong screening in one sample, a blind spot in another

The relation between rigid-force score and actual discrepancy differs sharply between samples (Figure 1). AIMNet2 has 263 records above 5 meV/Å and 14 above 10 meV/Å; none exceeds 25 meV/Å. Transition1x has 104 above 5 meV/Å but only one above 10 or 25 meV/Å. Its pooled metric is consequently sensitive to a single record.

At this budget, direct force/torque screening recovers 95.0% of the worst decile on AIMNet2, versus 87.8% for direct net-force screening and 28.04% expected under uniform selection (Table 1). Ridge and MP-pooled ridge both recover 77.6%, and log-error trees 76.0%. The learned policies do not improve on the strongest inexpensive check once seed cost is included.

Cost accounting changes the comparison. If seeds are already available, both ridge variants recover 77.6% including seed discoveries, while applying the rigid score only to the remaining

Policy	AIMNet2 (%)	Transition1x (%)
Random, direct	28.0 [25, 30]	28.0 [24, 35]
Net force, direct	87.8 [87, 88]	27.6 [23, 35]
Force/torque, direct	95.0 [95, 95]	31.8 [30, 35]
Force magnitude, direct	32.8 [32, 35]	30.6 [24, 38]
Geometry novelty, direct	68.6 [67, 72]	33.4 [28, 42]
Trees, log error	76.0 [72, 79]	39.0 [20, 49]
Trees, raw error	76.0 [72, 80]	36.4 [19, 48]
Trees, physical only	75.6 [71, 80]	29.2 [18, 39]
Ridge	77.6 [73, 83]	34.2 [19, 44]
MP-pooled ridge	77.6 [74, 81]	35.2 [22, 45]

Table 1: Worst-decile recall at matched total reference counts. Entries are scenario mean [minimum, maximum] in percent, not confidence intervals. Direct policies spend the entire budget on selection; learned policies pay seed costs. The average budget is 28.04% in each sample.

pool recovers 76.0%. Allowing the direct policy to spend the complete budget raises its result to 95.0%. The modest advantage conditional on existing seeds would be misleading as a claim about a new audit.

On Transition1x, direct force/torque screening recovers 31.8%, close to the random expectation of 28.04%. Log-error trees recover 39.0%, with a wide range of 20–49% over scenarios. MP-pooled ridge gives 35.2% and ordinary ridge 34.2%. These finite-sample differences do not establish a universal winner or a statistically demonstrated RMT advantage.

6.2 Detection count and discrepancy impact differ

The largest Transition1x discrepancy occurs at zero-based file index 255, formula C_6H_6O . Its frame RMSE is 866.48 meV/Å, while its net-force score is 0.3765 and rigid-component RMS 0.6691 meV/Å. It contributes 97.75% of total squared discrepancy. Both direct consistency policies miss it in every primary- budget scenario. This illustrates Proposition 1; it does not newly identify the mismatch’s electronic cause.

At matched total budgets, net-force and force/torque screening capture only 0.665% and 0.675% of Transition1x squared discrepancy. Ranking by original force magnitude captures 98.35%, despite recovering only 30.6% of the worst decile. Conversely, log-error trees recover more worst-decile records but capture 39.76% of squared discrepancy. Neither endpoint substitutes for the other (Figure 3).

Fixed tolerances further constrain interpretation. Every primary-budget direct net-force and rigid-force run recovers all 14 AIMNet2 records above 10 meV/Å. In Transition1x the only record above that threshold is the large mismatch: both direct consistency rules miss it, while force-magnitude ranking finds it. Recall at 25 meV/Å is undefined for AIMNet2 because no record exceeds the threshold; the result tables preserve that undefined value.

6.3 How many references should train the auditor?

An exploratory extension fixes the total cost at 280 references per 1,000-record sample. It uses the same five formula folds but draws nested random prefixes of 10, 20, 50 or 100 records from each designated seed fold, alongside the full fold (mean size 200). Five deterministic permutations are used per fold. The remaining four folds form a fixed audit pool, independent of prefix size. Unused seed-fold records are excluded from this pool. All seeds contain at least three formulas; the original grouped penalty selection and tree settings are retained.

A learned policy uses m references for training and $280 - m$ for acquisitions. Direct policies spend all 280 on the same pool. Only pool discoveries count, with positives defined as the worst $\lceil 0.1|\mathcal{U}| \rceil$ pool records. This transfer-to-unseen-formulas endpoint intentionally differs from

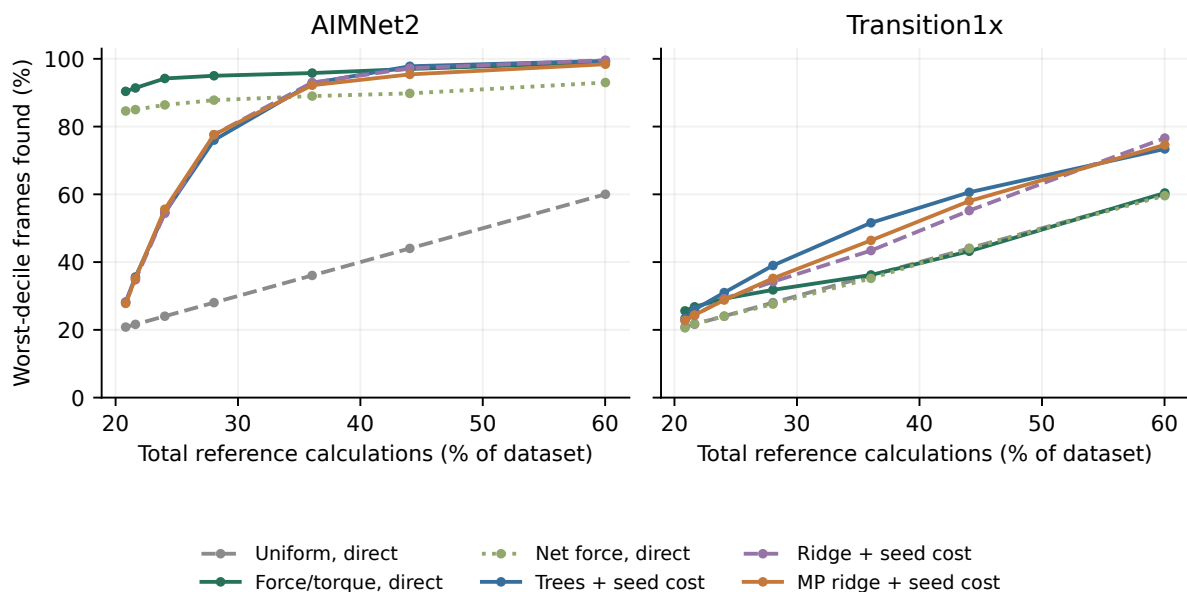


Figure 2: Recall versus total reference count. Points average five seed scenarios at each acquisition fraction. Direct policies receive the same per-scenario total budget as learned policies, including seed costs. Curves describe overlapping scenarios rather than independent replications.

the earlier whole-sample recall, which credited seed discoveries. Repeats are averaged within fold before summarizing the five overlapping scenarios; they are not independent confirmation datasets. The extension protocol was recorded after examining the primary results, and no fresh data were introduced.

With 50 seed references, log-error trees recover 92.8% of the pool’s worst decile on AIMNet2 and 36.5% on Transition1x. Using the entire seed fold gives 68.5% and 20.4%, respectively (Figure 4). At 50 references, ordinary ridge gives 93.7% and 36.5%, while MP-pooled ridge gives 88.0% and 35.3%. Direct force/torque screening gives 95.5% and 36.1% on these pools.

These curves measure the combined effects of predictor quality and reference allocation, not prediction error at equal acquisition counts. More training references need not produce more discoveries under a fixed total budget. The comparison does not establish a universal optimal seed size, an RMT advantage, or transfer from these molecular samples to a substantially larger database.

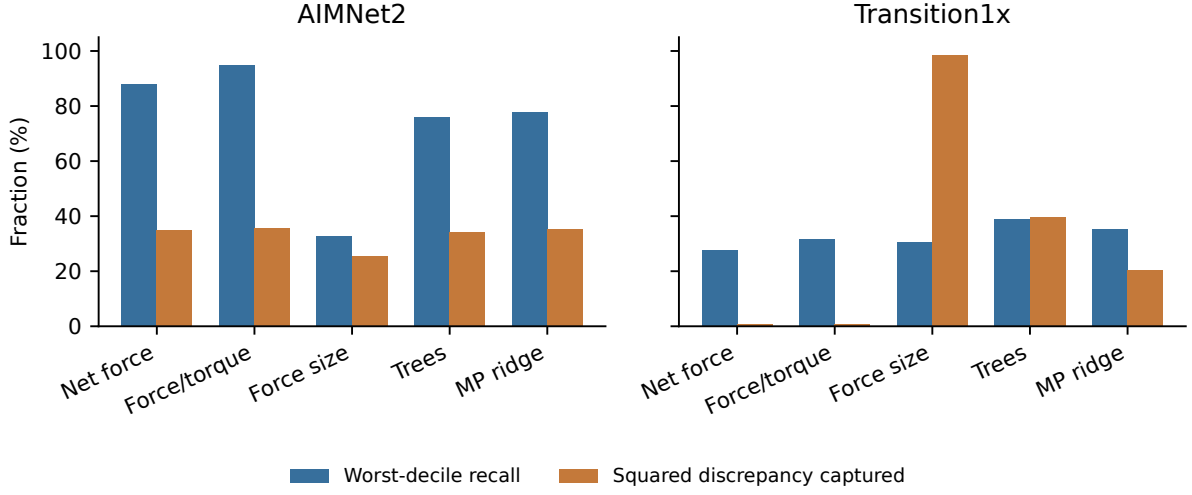


Figure 3: Detection count and squared-discrepancy impact at the primary matched budget. Transition1x impact is highly sensitive to whether its single large mismatch is audited. That record is retained in all primary results.

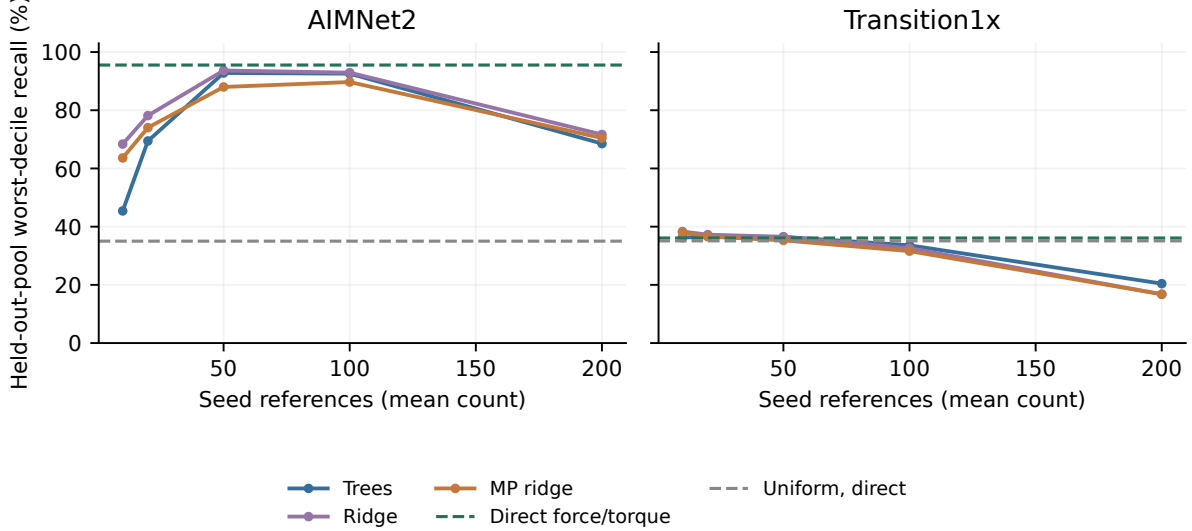


Figure 4: Reference allocation at a fixed total budget of 280. Increasing seed count leaves fewer acquisitions in the fixed held-out pool. Points average five seed permutations within each of five formula scenarios. The full-fold point has mean size 200; its actual size varies by scenario. Direct-policy lines use 280 pool acquisitions throughout.

7 Implications for large computational datasets

The main finding limits transfer of confidence from a diagnostic that works on one dataset to another. Consistency checks are inexpensive and reveal some discrepancies without training references. Yet an audit concentrating only on their violations can leave large symmetry-compatible discrepancies unchecked. Learned predictors add information but require reference-budget accounting and seed-sensitivity analysis.

For larger deployment, the proposed workflow retains provenance and solver diagnostics, runs inexpensive checks across all records, allocates an explicit initial reference budget, and evaluates complementary acquisition rules on held-out references. A portion can be reserved for random exploration beyond recognized failure patterns. The present experiment does not optimize this allocation or validate a hybrid policy; adding one after these outcomes requires a fresh evaluation protocol.

RMT can be tested for stabilizing correlations or examining repeated-calculation residuals. Our matrices are descriptor Gram matrices, whose bulk cannot be interpreted as a direct physical-noise spectrum. A small eigenvalue cannot be discarded merely because it looks statistically unremarkable. The demonstrated RMT variant is a comparison component, not a certificate of DFT accuracy.

Several limits govern extrapolation. This is a 2,000-configuration molecular experiment, not a hundred-thousand-record materials audit. It contains no periodic stress, magnetic-state or formation-energy ranking benchmark. The primary 20% average seed budget is substantial. The small-seed extension tests reference allocation within these samples, but transfer to larger databases remains untested. Reference counts ignore DFT cost variations with system size and composition. Pair-distance descriptors cost $O(n_i^2)$ per configuration, so the implementation does not establish linear scaling to large molecules or solids.

The operational references cannot isolate functional bias or all numerical causes. The samples also informed earlier exploratory work: the recorded protocol fixes choices for this audit but does not erase that history. Overlapping scenarios provide sensitivity analysis rather than independent statistical replication. Further evidence should use fresh paired data, cost-weighted evaluation and downstream discovery decisions.

8 Conclusion

DFT data reliability can be studied as a budgeted acquisition problem with explicit reference costs and distinct detection-count and error-impact objectives. Our paired-data experiment shows strong inexpensive screening on AIMNet2 and a severe blind spot on Transition1x. Seed costs change the comparison with learned auditors, while the chosen utility changes which policy looks useful. The results motivate reporting diagnostic limitations alongside successes. They do not establish universal noise removal, a special RMT advantage or improved discovery outcomes.

Data and code availability

Original paired files are provided by the authors of Kuryla et al. [2025] at https://github.com/water-ice-group/datasets_dft_accuracy. The accompanying archive contains the complete analysis, recorded audit protocol, immutable download manifest, per-scenario predictions and outcomes, and manuscript source. Original files are downloaded rather than redistributed. Analysis code, results and submission files are available at https://github.com/SinanGokmen/rmt-reaction-kinetics/tree/main/dft_reliability.

Generative AI assistance

ChatGPT/Codex was used extensively in formulating the research question and mathematical arguments, implementing and checking code, running the analyses, searching the literature, and preparing the manuscript and figures. Numerical results were produced by the accompanying scripts.

A Reproducibility and supporting checks

Descriptors and regression. Element channels are H, C, N, O, F, S, Cl and other. For each per-atom or pairwise array we use minimum, 25th percentile, median, 75th percentile, maximum and standard deviation, followed by $\log(1 + x)$. Net, rotational and rigid scores use $\log(1 + s/10^{-3})$ with forces in eV/Å. Charge is untransformed and atom count uses $\log(1 + n)$. ExtraTrees uses 256 trees, minimum leaf size five, all features available per split, and random seed 20260911 + s for scenario s . Its physical ablation uses only net-force and rigid-force inputs. Log targets are $\log \max(q, 10^{-8})$, with q in eV/Å.

Ridge standardizes nonconstant descriptors using training means and standard deviations, dividing the Gram matrix by training count. Penalties 10^{-4} , 10^{-2} , 1, 100 are selected by mean log-target squared error in grouped three-fold validation within seeds. The MP variant replaces eigenvalues below the nominal edge by their mean before adding the ridge penalty. It is independently tuned and refitted on all seed references.

Novelty and splits. Novelty is squared standardized distance from the whole sample’s median in geometry, composition, charge and atom-count descriptors, excluding force-derived inputs. Standard deviations use the unlabeled whole sample and are floored at 10^{-8} . A formula’s fold is the integer represented by the first 12 hexadecimal digits of SHA-256 of `dft-denoise-v1`: concatenated with the sorted formula, modulo five. Ranking ties use increasing original file index. Each primary worst-decile set contains exactly 100 records.

Unresolved file conventions. The available ANI-1x file yields 173.86 meV/Å after the attempted Hartree-to-eV conversion, rather than the reported comparison. The SPICE file contains 998 frames, a gradient-sign convention and a separate D3 array; the attempted comparison yields 16.68 meV/Å. Their correspondence to the published comparisons remains unresolved. Neither is used in the fitted audit. These checks do not identify an error in the source publication.

Supporting denoising analysis. The earlier pilot, preserved in the archive, projects discrepancies into allowed and rigid subspaces. The allowed component contains 92.28% of squared discrepancy on AIMNet2 and 99.89% on Transition1x. Element-weighted correction reduced AIMNet2 label RMSE from 4.7979 to 4.4800 meV/Å in its separate grouped protocol. These are corrected-label results, not the present detection metrics or improved potential-training results.

Verification. Changing tight forces and energies while holding original results fixed changes no input feature. Rigid coordinate transformations change descriptors by at most 5.63×10^{-14} on checked configurations. Saved predictions never pair a target with its seed formula fold. Primary recalls reconstructed from saved scores match exactly, and acquisition sets have the specified distinct-record counts. The archive includes 80,000 scenario-method score rows and 4,760 metric rows for 2,000 unique configurations; those rows are not additional independent samples. The small-seed extension adds 250 seed sets, 1,000,000 score rows and 2,250 metric rows on the same configurations. Independently reconstructing its acquisition sets and metrics agrees to within 1.12×10^{-16} ; all nested-seed and formula-separation checks pass.

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